End-to-End Performance Modelling and Analysis

Communication Networks • Continuous Assessment 1
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A reproducible model, with explicit assumptions and measured uncertainty.

Three tasks, one image journey

01

Graph model
Link attributes and route selection

02

Wireless access
Shannon rate and polling delay

03

Queueing model
M/M/1 and the arrival-rate limit

The integrated simulation checks whether the assumptions remain valid when the stages are connected.

An illustrative, fully declared baseline

4 cameras

Independent Poisson image arrivals

1 Mbit

Fixed image size in baseline

20 MHz

Scheduled bandwidth; SNR = 10 dB

50 images/s

MEC mean service rate

100 ms

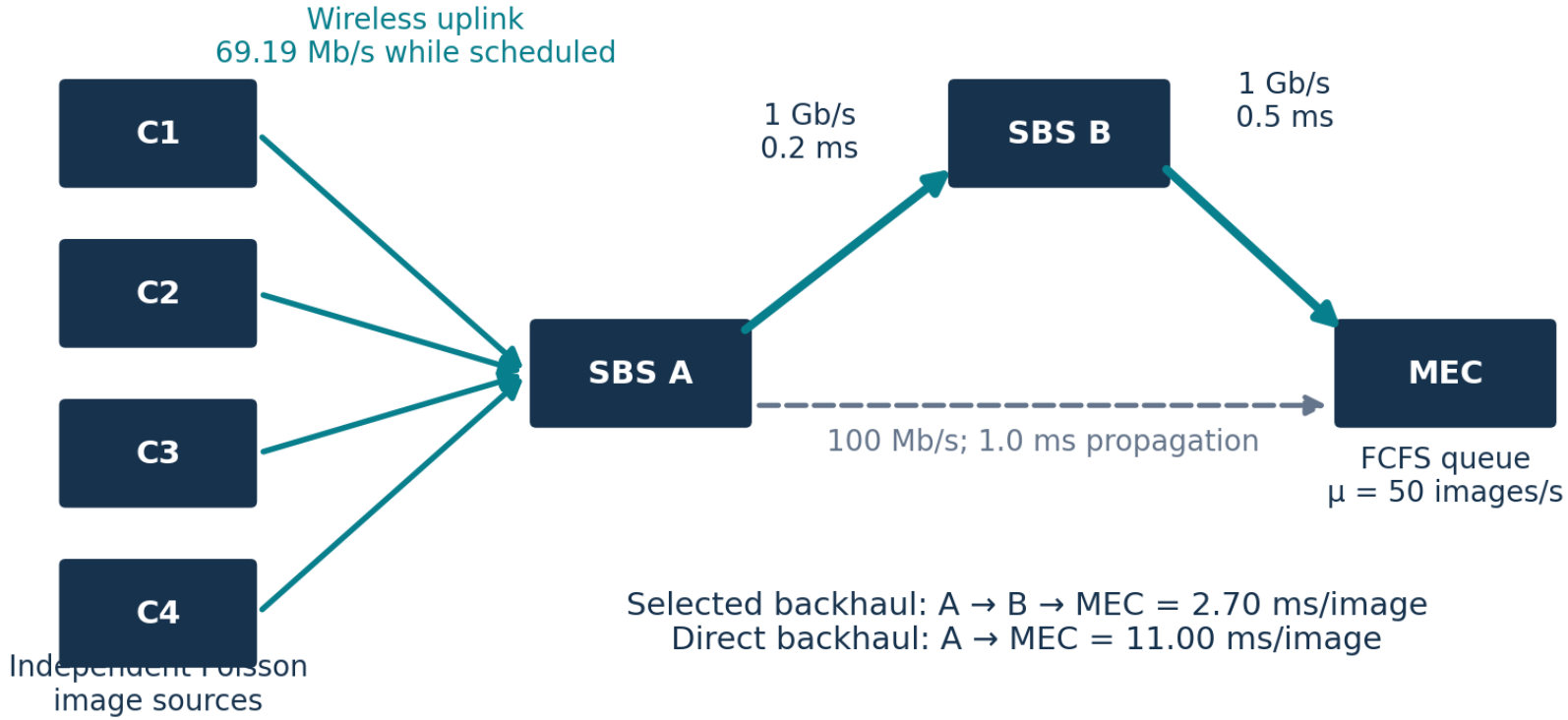
Chosen mean-delay target

0.20 ms

Grant / guard overhead per visit

A graph makes the routing decision explicit

Figure 1. Directed topology and the selected store-and-forward route



Link weight = image serialization time + propagation delay. Source: report Section 3.

The extra hop is faster in this graph

11.00 ms

Direct wired backhaul: A → MEC

2.70 ms

Selected wired backhaul: A → B → MEC

Selected path: C1 → A → B → MEC
Total route weight: 17.15 ms; saving: 8.30 ms

Shannon gives the scheduled radio rate

$$R = W \log_2(1 + SNR_{linear})$$

69.19 Mbit/s

Ideal payload rate while scheduled

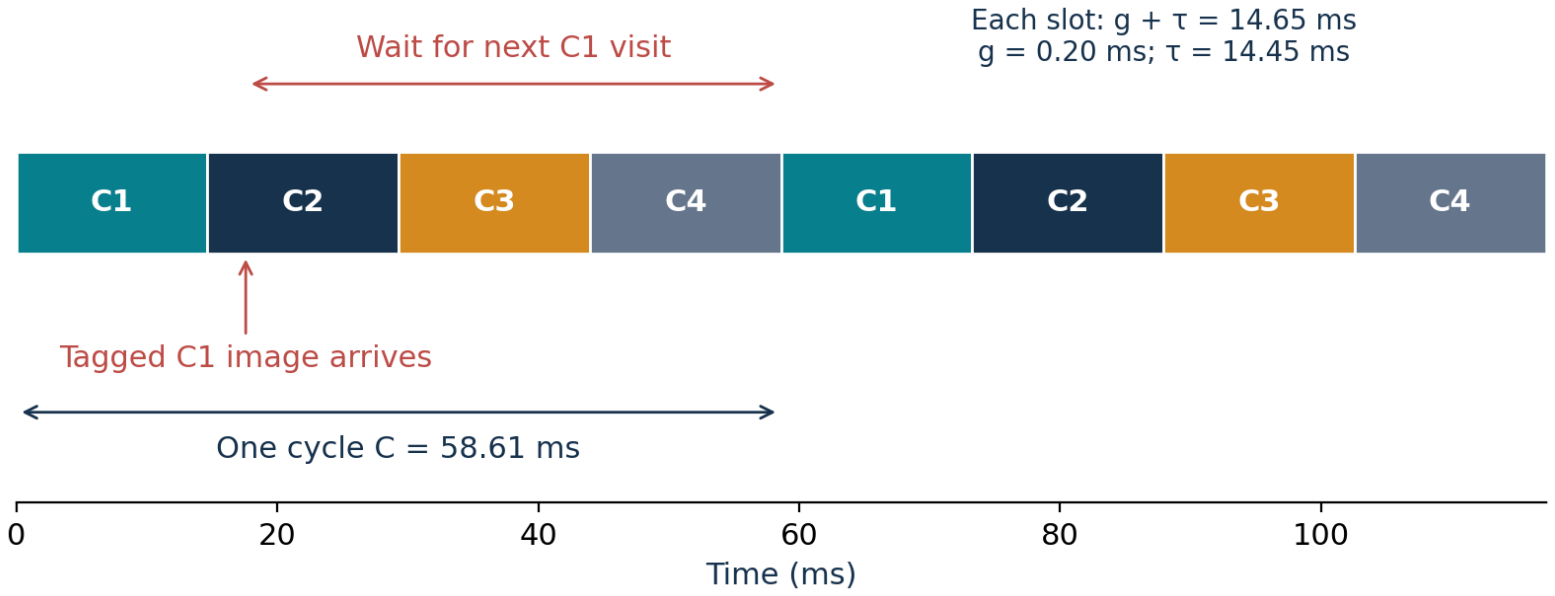
14.45 ms

Transmission time $\tau = S / R$

OFDMA resources are assigned in time; bandwidth is never counted four times.

A visit is not immediately available

Figure 2. Reserved cyclic polling; one image per camera visit



C = N(g + τ) = 58.61 ms

Tagged mean wireless time = C/2 + g + τ = 43.96 ms

Camera backlog adds another load effect

Mean polling wait = C / [2(1 − r)]

$r = (\lambda / N)C = \lambda(g + \tau)$
Radio stability: $\lambda < 68.24$ images/s

46.25 ms

Mean polling wait at $\lambda = 25$/s

70.81 ms

Mean polling wait at $\lambda = 40$/s

REPRODUCIBLE SIMULATION

Test each model, then connect the stages

30

Independent replications per rate

100,000

Measured images per replication

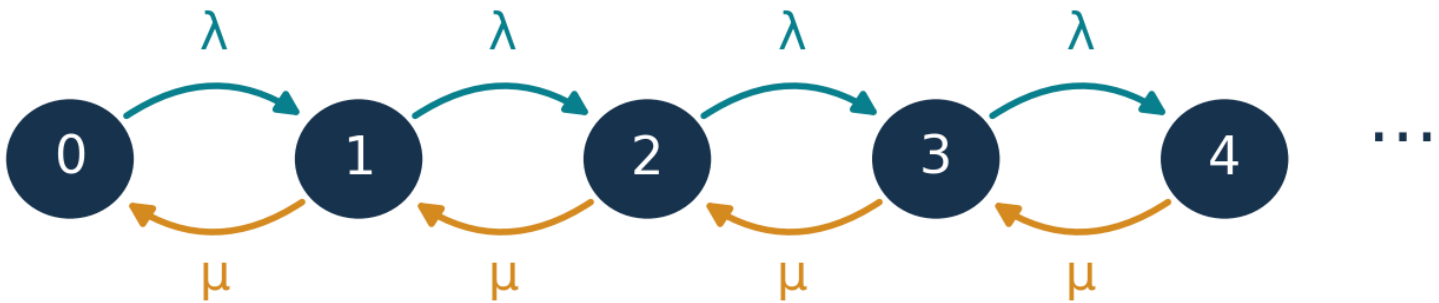
20,000

Warm-up images discarded

11 arrival rates • Fixed published seed • 95% confidence intervals
10 deterministic tests passed; raw replication results included.

The required M/M/1 mean-delay bound

Figure 3. M/M/1 birth–death chain (no departure from state 0)

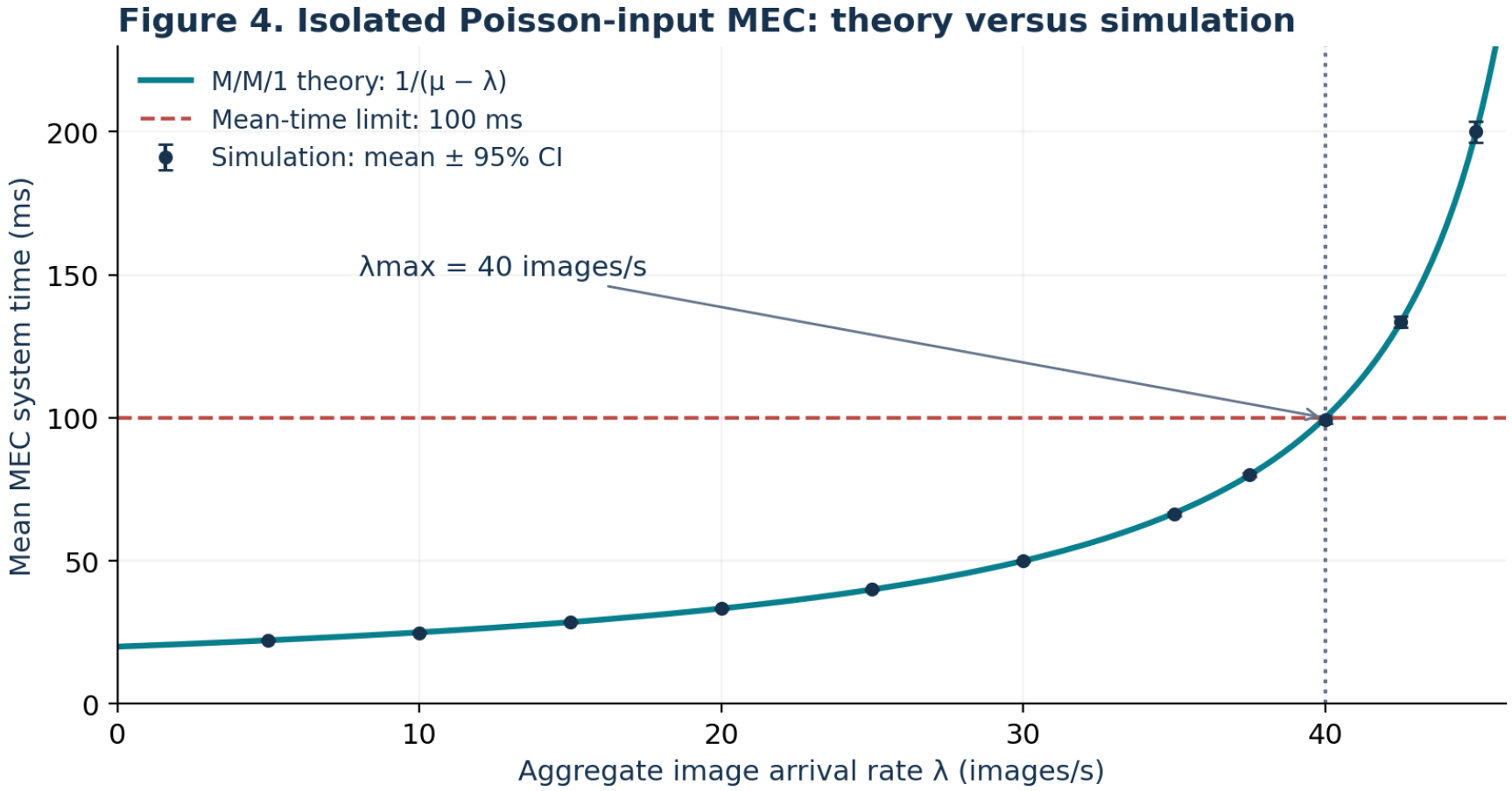


State k = images waiting + the image being processed

$$T = W_s = W_q + 1/\mu = 1/(\mu - \lambda)$$

$\lambda_{max} = \mu - 1/T_{max} = 40$ images/s
Feasibility: $T_{max} > 1/\mu$; stability: $\lambda < \mu$.

Theory and simulation agree at the server



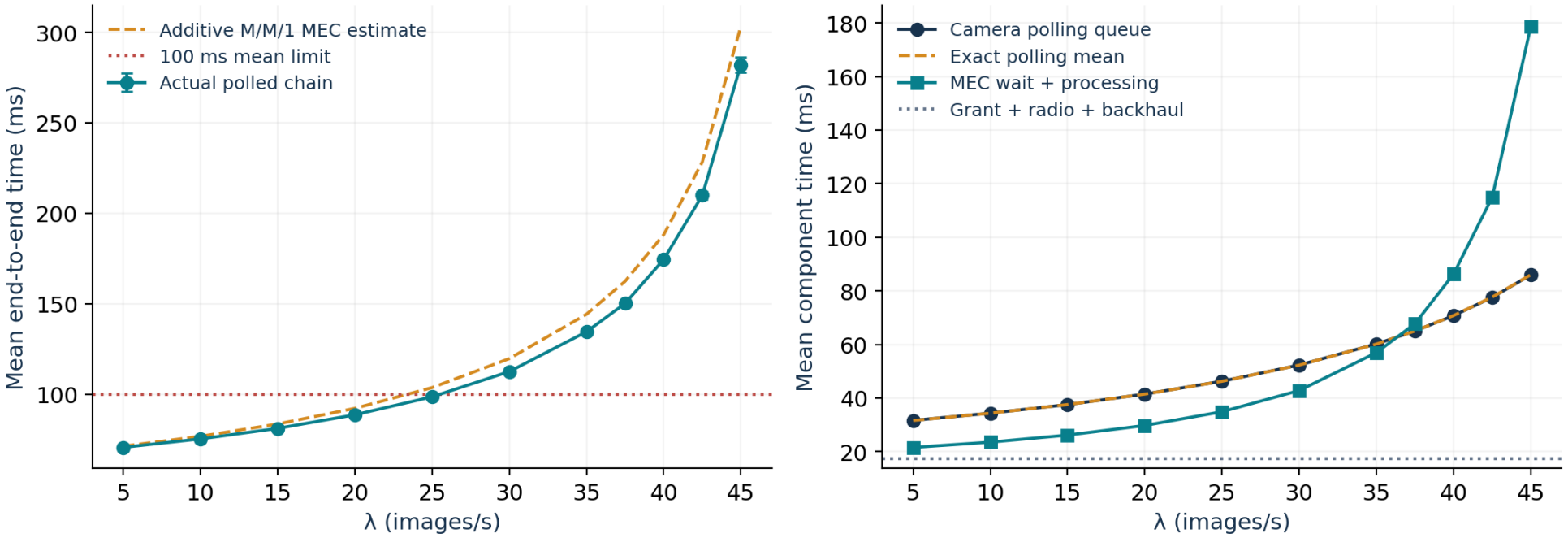
0.62%

Maximum relative difference in the mean

10 of 11 pointwise intervals cover theory.

The complete chain has a lower latency capacity

Figure 5. Actual cyclic polling changes MEC arrival statistics



25 images/s: 98.52 ms | 30 images/s: 112.48 ms

Upgrade radio and processing together



Combined upgrade: 40 MHz + 75 images/s MEC; deadline misses remain 9.51%.

Design for the whole journey

01 Choose routes using capacity and delay.

02 Include both polling backlog and MEC queueing.

03 Check mean delay and deadline reliability separately.

Questions

Sources: assessment brief; course Chapter 2; Shannon (1948); Dijkstra (1959); Little (1961). Full references and derivations are in the report.